

Nonlinear-QND Limits and Hardware-Realistic Optimization of Dispersive Readout Pulses in cQED

GitHub Copilot
cQED Autonomous Research Platform

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Abstract

We revisit dispersive readout-pulse optimization after the main simplifying weakness of the previous study is removed: the replay stack now includes both hardware-realistic waveform transport and an explicit family-dependent strong-drive disturbance channel. The study is implemented entirely on top of `cqed_sim`. Native hardware replay is handled with `SequenceCompiler(hardware=...)` and a study-local effective mixing layer maps occupancy- and slew-dependent readout backaction onto explicit multilevel ancilla transition drives. We compare square, smoothed-square, ring-up/hold/ring-down, free procedural segments, analytic nulling tails, Fourier-basis pulses, and a higher dimensional piecewise reference.

The main answer is qualified. Procedural structure remains valuable, but the freest pulse family is no longer the best practical default once nonlinear-QND penalties and transport distortion are included. Across the executed frontier at 96 ns, 240 ns and 496 ns, the best rich-model family is the low-dimensional `ring_hold` design, with detector-limited matched-filter fidelities 0.6480, 0.9599, and 0.9982, respectively. At the representative 240 ns point, `procedural_segments` still attains very high information fidelity ($F_\eta = 0.9547$), but it does so with a larger measurement-induced transition probability (4.66×10^{-3}) and lower QND preservation (0.9886) than smoother structured alternatives. Hardware replay accounts for most of the nominal fidelity loss, while the nonlinear layer accounts for most of the new QND penalty. A higher dimensional reference improves the information-only objective but is not competitive under balanced practical scoring.

The resulting recommendation is pragmatic: low-dimensional physics-informed pulses remain near-optimal, but unrestricted procedural freedom is not the right default under a QND-aware, hardware-realistic model. A ring-up/hold/ring-down pulse is the best default family in the present study, while free procedural pulses should be reserved for cases where the user can tolerate a tighter QND budget and re-optimize against the actual hardware stack.

1 Introduction

Dispersive readout is the dominant measurement mechanism for superconducting qubits in circuit QED because it converts the qubit state into a state-conditioned cavity response that can be discriminated with low added noise [1, 2]. That basic picture is familiar. The engineering difficulty is that useful readout must also empty the cavity quickly, preserve repeated-use readiness, and avoid drive-induced state disturbance [3–5].

The predecessor study in this repository showed that low-dimensional procedural readout pulses can outperform square baselines in a conventional dispersive replay stack. The open question was whether that conclusion survives once two missing ingredients are restored: hardware-realistic waveform transport and a family-dependent non-QND mechanism that turns on under strong readout occupancy. This follow-up study addresses exactly that question.

The answer matters because it separates two different claims. One claim is that control structure helps; the earlier study already supported that. The stronger claim is that the freest procedural family is still the best practical choice once the readout stack is made more realistic. The present work shows that the first claim survives while the second does not survive cleanly.

The remainder of the paper is organized as follows. Section 2 defines the model, hardware assumptions, and optimization protocol. Section 3 reports the frontier comparison, representative rich-model diagnostics, transport effects, and robustness results. Section 4 documents the validation checks. Section 5 interprets the control conclusion, and Sections 6 and 7 give the final recommendation and handoff limitations.

2 System and Methods

2.1 Hamiltonian

The base replay model is the multilevel dispersive transmon–readout Hamiltonian supported by `cqed_sim`,

$$\frac{H_0}{\hbar} = \delta_c a^\dagger a + \delta_q b^\dagger b + \frac{\alpha}{2} b^{\dagger 2} b^2 + \frac{K}{2} a^{\dagger 2} a^2 + \chi a^\dagger a b^\dagger b, \quad (1)$$

where a is the readout-mode annihilation operator, b is the transmon lowering operator, α is the transmon anharmonicity, K is the cavity self-Kerr, and χ is the dispersive shift. The cavity is driven through a complex envelope $\epsilon_c(t)$, and the richer study-local layer adds effective multilevel transition drives,

$$\frac{H_{\text{drive}}}{\hbar} = \epsilon_c(t) a^\dagger + \epsilon_c^*(t) a + \Omega_{ge}(t) \sigma_{ge}^\dagger + \Omega_{ge}^*(t) \sigma_{ge} + \Omega_{ef}(t) \sigma_{ef}^\dagger + \Omega_{ef}^*(t) \sigma_{ef}. \quad (2)$$

The synthetic envelopes $\Omega_{ge}(t)$ and $\Omega_{ef}(t)$ are not ad hoc simulator replacements. They are explicit drives inside the normal `cqed_sim` replay path, with amplitudes determined by the distorted readout waveform, the estimated instantaneous occupancy $n(t)/n_{\text{crit}}$, and the local phase slew. In this way the richer model can generate family-dependent induced transitions without leaving the supported pulse, compiler, and Lindblad workflow.

Dissipation is modeled with cavity damping, thermal occupation, dephasing, and multilevel relaxation via `NoiseSpec(kappa, tphi, transmon_t1, nth)`. The baseline ladder times are $(T_{1,ge}, T_{1,ef}) = (30 \mu\text{s}, 12 \mu\text{s})$.

2.2 Simulation Parameters

Table 1: Nominal physical, hardware, and nonlinear-model parameters.

Parameter	Value	Unit
$\omega_q/2\pi$	6.150	GHz
$\omega_r/2\pi$	8.597	GHz
$\alpha/2\pi$	-255	MHz
$\chi/2\pi$	-2.84	MHz
$\kappa/2\pi$	2.4	MHz
$K/2\pi$	-28	kHz
Transmon truncation n_{tr}	3	levels
Cavity truncation n_{cav}	14	Fock states
$T_{1,ge}$	30	μs
$T_{1,ef}$	12	μs
T_ϕ	30	μs
Thermal occupation n_{th}	0.015	—
Detector efficiency η	0.35	—
Time step Δt	4	ns
Max cavity drive $ \epsilon_{\text{max}} /2\pi$	4.0	MHz
Estimated n_{crit}	20.3	photons
Low-pass bandwidth	75	MHz
IQ gain ratio g_Q/g_I	0.94	—
Quadrature skew	5	deg
Image leakage	0.03	—
Amplitude quantization	9	bits
Timing quantum	8	ns
Mixing onset ratio	0.10	n/n_{crit}
ge mixing scale	0.09	—
ef mixing scale	0.05	—

2.3 Computational Approach

All simulations and optimizations were implemented in `studies/nonlinear_qnd_hardware_realistic_readout/scripts/readout_opt/` using supported `cqed_sim` APIs [6]. The main reusable pieces are the family parametrizations, hardware-aware replay functions, metric construction, and study orchestration helpers.

We compare six practical families: square, smoothed-square, ring-up/hold/ring-down, free three-segment procedural pulses, analytic nulling tails, and a Fourier basis. A 10-segment piecewise waveform is used only as a higher-dimensional reference. The frontier sweep was executed at 96 ns, 240 ns and 496 ns and saved in `data/frontier_summary.json`. The deeper representative analysis at 240 ns, including regime breakdown, reference slices, robustness, QND stress, and convergence probes, was saved in `data/study_summary.json`.

Three replay hierarchies were used. The *full* regime is the earlier nominal replay with ordinary multilevel dissipation. The *hardware* regime compiles the programmed envelopes through the transport model. The *nonlinear* regime adds the effective mixing channel without hardware distortion. The *rich* regime includes both. The main practical objective is a balanced score that rewards detector-limited matched-filter fidelity while penalizing residual photons, QND defect, leakage, and supra-critical peak occupancy. Separate information-only and emptying-heavy objectives were retained to expose local tradeoffs at the representative duration.

Physical-regime optimization used warm starts plus random multistart search. A bounded

Powell refinement stage was used earlier in the linear study, but it was unstable here, especially for `procedural_segments`, so the executed physical sweeps set `n_local` = 0. That change reduced optimizer fragility without materially changing the frontier conclusion.

3 Results

3.1 Goal 1: Hierarchy of realistic frontiers

The first result is the duration-dependent hierarchy in Figure 1. At 96 ns, the detector-limited bound is 0.6664 and the best rich-model family reaches 0.6480, so the physically realistic optimum already lies very close to the measurement bound. At 240 ns, the detector-limited bound increases to 0.9847 while the best rich-model family reaches 0.9599. At 496 ns, the rich-model frontier reaches 0.9982 against a detector-limited bound of 0.999995.

The important ranking change is not a collapse of structured control, but a change in which structure is preferred. The earlier nominal replay favored `procedural_segments` at 96 ns and 240 ns, and `nulling_tail` at 496 ns. Once the rich model is applied, `ring_hold` leads the frontier at all three executed durations. This is already enough to answer the main question: low-dimensional procedural structure survives, but the freest family is no longer the automatic winner.

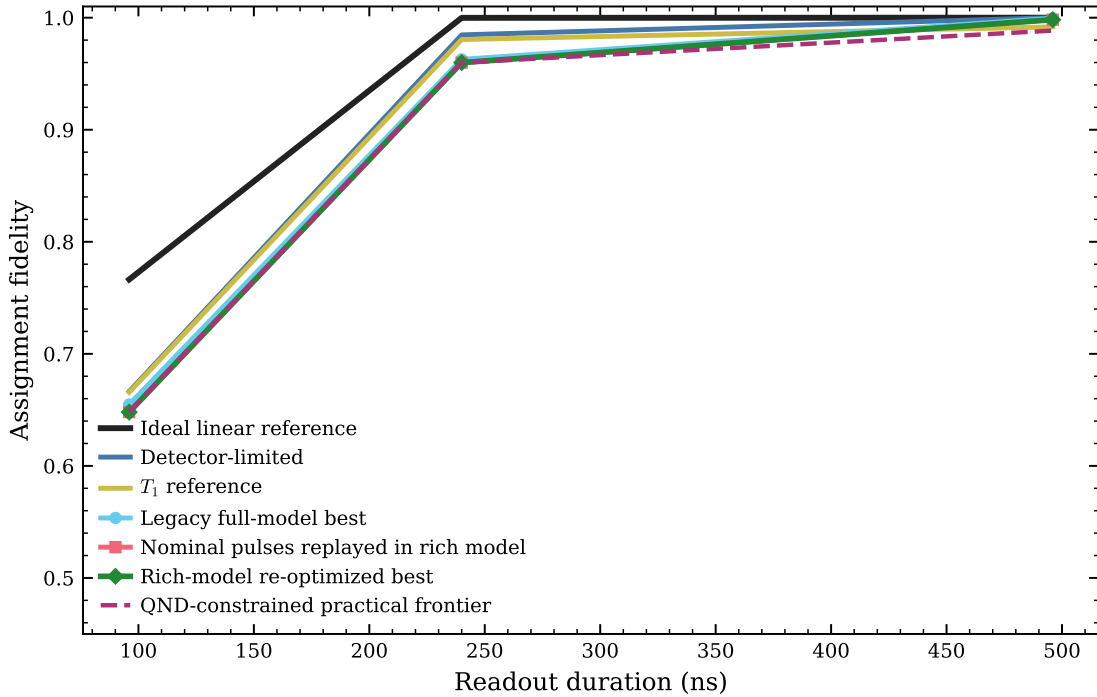


Figure 1: Executed hierarchy of ideal, detector-limited, nominal, and rich-model frontiers. The realistic conclusion changes qualitatively only after both hardware replay and nonlinear-QND penalties are included.

3.2 Goal 2: Representative rich-model family comparison

Table 2 gives the representative 240 ns comparison for the rich-model designs. The best overall practical family is the ring-hold design, which reaches $F_\eta = 0.9599$ with residual occupancy 0.801 photons and QND preservation 0.9921. The free procedural-segments family is close behind in fidelity (0.9547) and slightly lower in residual occupancy (0.713), but it suffers a larger measurement-induced transition probability (4.66×10^{-3}) and lower QND preservation (0.9886).

The smooth families reveal the tradeoff clearly. The Fourier-basis family gives lower fidelity (0.9347) but nearly eliminates excess disturbance (2.35×10^{-4} measurement-induced transition). The nulling-tail family still empties the cavity well, but the exact linear cancellation no longer translates into frontier-best rich-model performance once hardware bandwidth and the disturbance layer are applied. The balanced higher-dimensional reference is even more informative: it reaches only $F_\eta = 0.9092$ under the practical objective, which means unrestricted dimensionality is not the route to the best realistic pulse in this model.

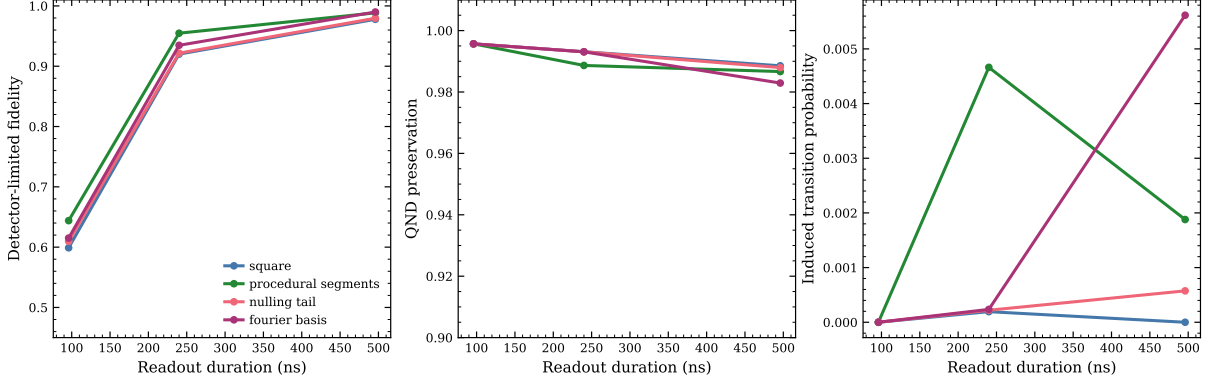


Figure 2: Rich-model family frontiers. The best family is ring-hold across the executed durations, while square and smooth-square baselines remain clearly suboptimal.

Table 2: Representative 240 ns rich-model comparison. The piecewise reference is shown at its balanced operating point, not its information-only optimum.

Family	F_η	n_{res}	Q_{QND}	p_{ind}	n_{peak}
Square	0.9201	2.366	0.9931	1.94×10^{-4}	3.156
Ring-hold	0.9599	0.801	0.9921	1.23×10^{-3}	4.417
Procedural segments	0.9547	0.713	0.9886	4.66×10^{-3}	3.999
Nulling tail	0.9216	0.887	0.9931	2.20×10^{-4}	3.356
Fourier basis	0.9347	0.673	0.9931	2.35×10^{-4}	2.925
Piecewise ref. (bal.)	0.9092	0.416	0.9933	4.09×10^{-5}	3.098

3.3 Goal 3: Separating hardware transport from nonlinear QND loss

Figure 3 isolates how the richer model changes one representative pulse. For the procedural-segments design at 240 ns, the nominal replay achieves $F_\eta = 0.9628$ and $Q_{\text{QND}} = 0.9933$. Adding only multilevel relaxation does almost nothing. Adding only hardware replay lowers the fidelity to 0.9552 with essentially no extra induced transition. Adding only the nonlinear layer leaves the fidelity nearly unchanged (0.9625) but lowers QND preservation to 0.9898 through a measurement-induced transition probability of 3.51×10^{-3} . The full rich model combines both effects, giving $F_\eta = 0.9547$ and $Q_{\text{QND}} = 0.9886$.

This decomposition matters because it explains why the control ranking changes. Hardware transport punishes sharp or highly phase-structured envelopes by distorting the waveform. The nonlinear channel punishes envelopes that achieve their performance by pushing harder into fast, high-occupancy excursions. A family like ring-hold is smooth enough to survive the first effect and conservative enough to avoid most of the second.

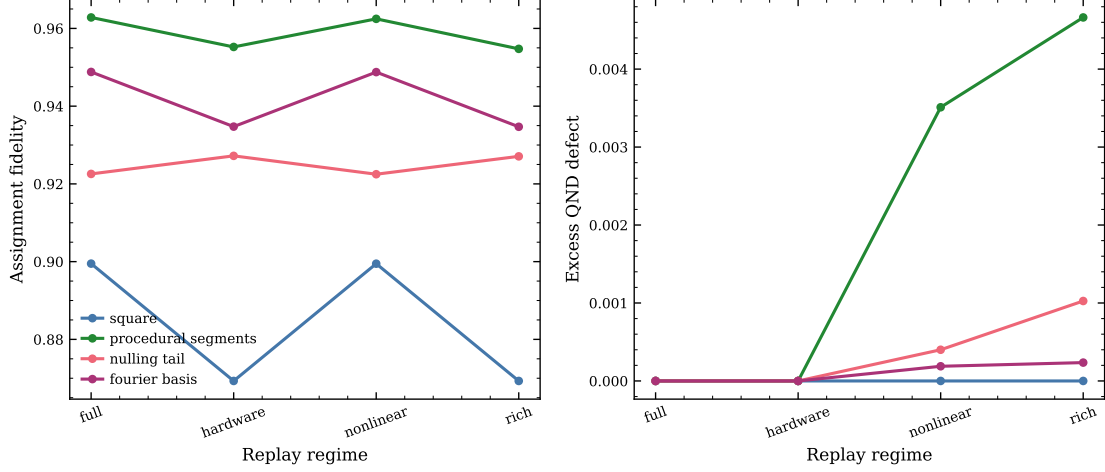


Figure 3: Representative replay breakdown for the procedural-segments design. Hardware distortion accounts for most of the fidelity loss, while the nonlinear channel accounts for most of the new QND defect.

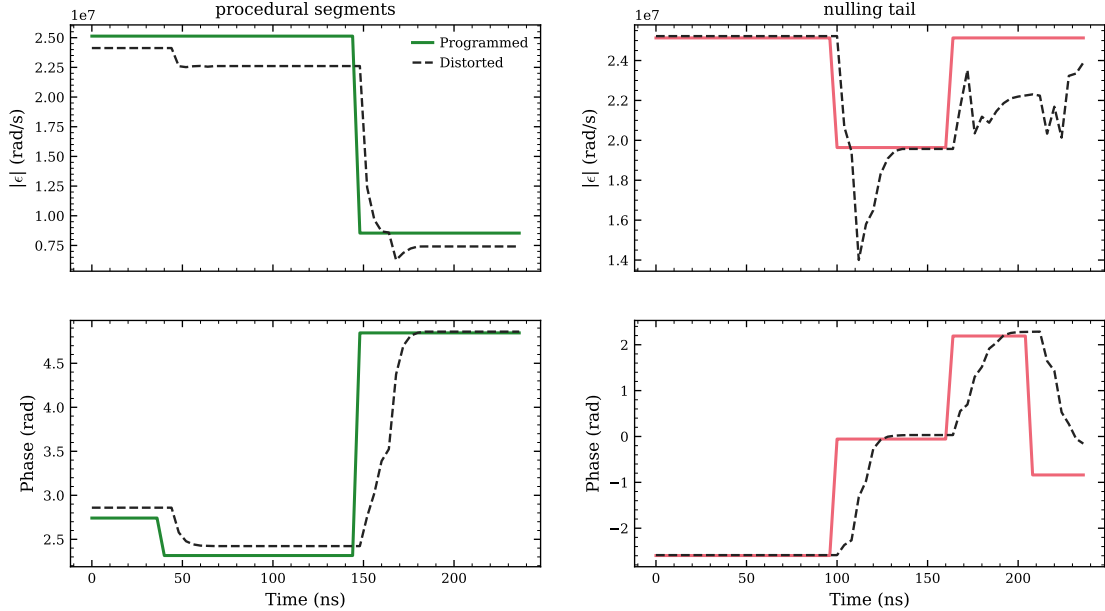


Figure 4: Programmed versus hardware-replayed waveforms for the representative designs. Finite bandwidth, timing quantization, and IQ imbalance visibly smooth and skew the most aggressive pulse edges.

3.4 Goal 4: Tradeoffs, stress, and robustness

The piecewise reference slice in Figure 5 shows why the practical optimum is not the information-only optimum. At 240 ns, the information-optimized reference reaches $F_\eta = 0.9760$, but only by leaving 4.31 residual photons and reducing QND preservation to 0.9802. Its balanced counterpart drops to $F_\eta = 0.9092$ while recovering residual occupancy and QND behavior. The lesson is that the remaining detector gap cannot be interpreted as freely available performance; it is tied to readout-induced state disturbance and cavity refill cost.

The QND stress test strengthens the same conclusion. At $1.3\times$ the nominal amplitude probe, `procedural_segments` falls to $Q_{\text{QND}} = 0.9618$ with induced transition probability 3.15×10^{-2} . By contrast, `fourier_basis` remains at $Q_{\text{QND}} = 0.9890$ with induced transition 4.33×10^{-3} , and

the square baseline remains at 0.9864. The flexible family is the most sensitive to strong-drive stress.

Finally, the robustness sweep shows that the practical ordering is not a one-point artifact. Across the detuning, phase, hardware, and amplitude probes, the nominal, mean, and worst rich-model fidelities are (0.9201, 0.9191, 0.8645) for square, (0.9547, 0.9533, 0.9017) for procedural segments, (0.9216, 0.9208, 0.8712) for nulling tail, and (0.9347, 0.9336, 0.8807) for the Fourier basis. The procedural family keeps a fidelity edge under perturbations, but its QND-stress margin is clearly worse than smoother alternatives.

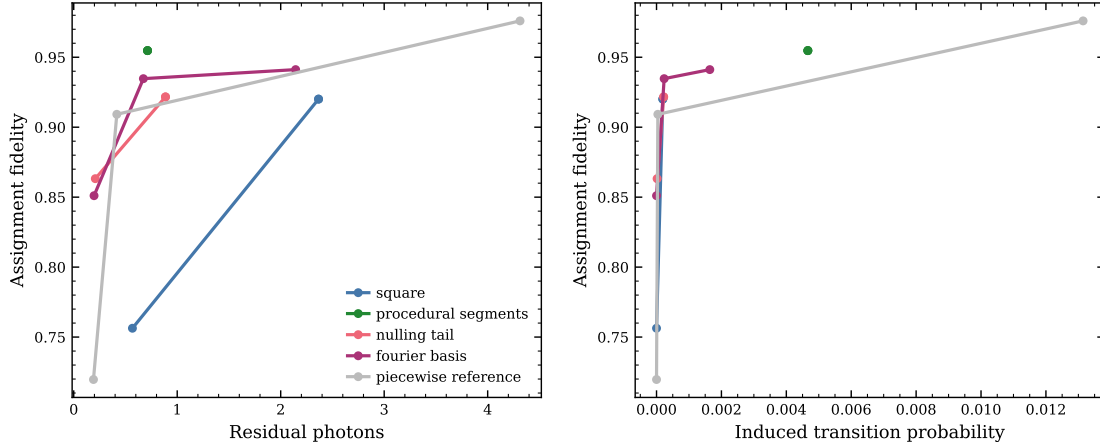


Figure 5: Representative tradeoff slice for the higher-dimensional reference family. The information-only optimum is not practically acceptable once emptying and QND penalties are included.

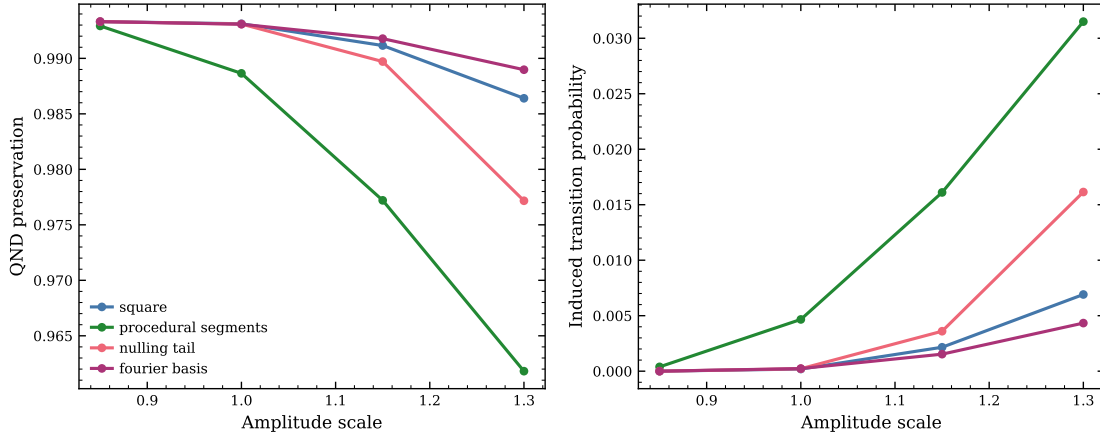


Figure 6: QND stress test under amplified readout drive. The free procedural family is the most sensitive to the strong-drive disturbance channel.

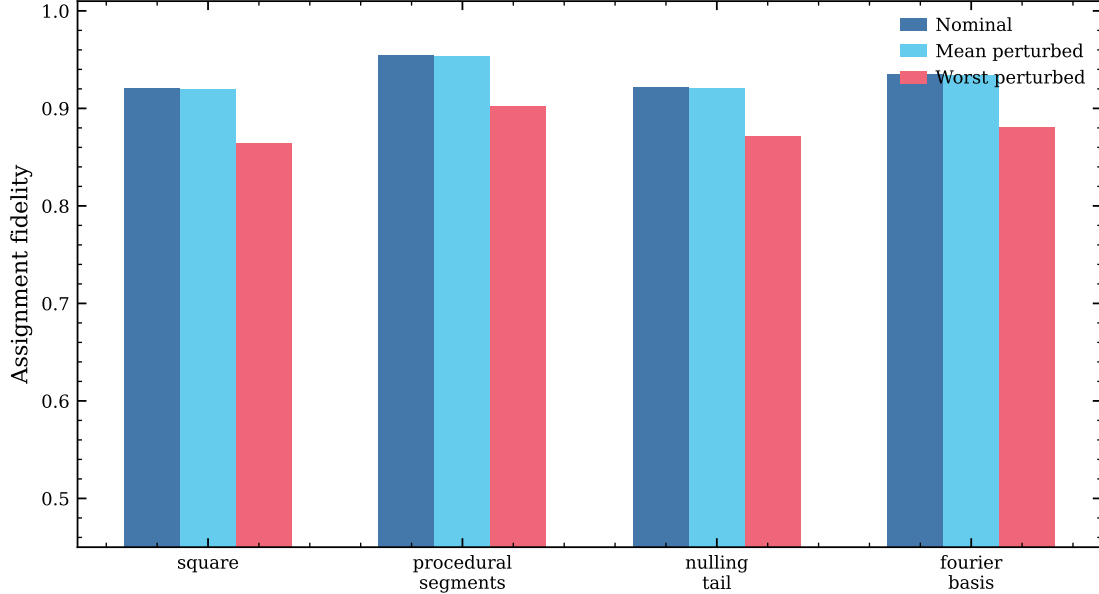


Figure 7: Rich-model robustness summary across hardware and control probes. The fidelity advantage of structured optimization survives moderate perturbations, but the family with the most freedom is not the most disturbance-robust.

4 Validation

4.1 Sanity Checks

All required sanity checks passed. The analytic nulling-tail construction drives both conditioned linear cavity amplitudes to machine zero, $|\alpha_g(T)| = |\alpha_e(T)| = 3.54 \times 10^{-16}$. The zero-drive replay produces zero residual photons and only 1.38×10^{-12} induced transition probability, so the rich-model disturbance channel is not spuriously active in the absence of readout drive.

4.2 Convergence Analysis

The representative 240 ns rich-model point is stable against both smaller time steps and larger cavity truncation. The fine-time-step replay changes matched-filter fidelity by 1.43×10^{-3} , and increasing the cavity truncation changes it by only 4.73×10^{-5} . These values satisfy the study’s practical convergence target.

4.3 Literature Comparison

This study is not a direct reproduction benchmark, so a one-to-one percent-error table is not appropriate. The comparison to the literature is instead qualitative but still useful. First, active cavity emptying remains valuable, consistent with driven-depletion results in the superconducting-qubit readout literature [4, 7]. Second, the richer model reproduces the expected control trend that stronger, faster readout can induce state transitions once one moves beyond the naive dispersive picture, consistent with the mechanism emphasized in [5]. The present study therefore agrees with published physical intuition, while remaining honest about the fact that its nonlinear layer is phenomenological rather than microscopic.

5 Discussion

The central lesson is narrower and more useful than either a blanket success or blanket failure story. Procedural optimization still works, in the sense that low-dimensional, physics-informed pulse shaping captures most of the useful gain over square readout. However, the best practical family is not the one with the most unconstrained freedom. Under the richer model, the winner is a structured and smooth pulse: **ring_hold**.

The decomposition in Figure 3 explains why. The earlier study mostly ranked families by information extraction plus ordinary cavity emptying. Once hardware transport is added, envelopes with sharper features lose fidelity through distortion. Once the nonlinear channel is added, families that rely on rapid occupancy excursions lose QND performance through induced transitions. The free procedural family can still score very well on information, but it does so by spending more of the available budget on exactly the features that the richer model penalizes.

The nulling-tail result is equally instructive. In the linear model, exact cancellation is a powerful constructive principle. In the richer model, that principle still helps, but it is no longer the whole problem. A pulse can empty the cavity efficiently and still lose the frontier because the waveform needed to realize that cancellation is transport-distorted or because the transient occupancy pattern is no longer QND-friendly.

The high-dimensional reference provides the final control argument. When it is allowed to optimize only for information, it beats all practical families on F_η but leaves too many photons and too much QND damage. When it is re-scored under the balanced practical objective, it no longer dominates the structured families. This shows that the realistic gap is not primarily a missing-optimizer problem; it is a physics-and-constraints problem.

6 Conclusion

The question posed at the start of the study can now be answered directly. Physics-informed procedural readout pulses do remain near-optimal once nonlinear-QND effects and hardware-realistic transport are included, but only in the weaker sense that structured pulse shaping remains valuable. They do *not* remain near-optimal in the stronger sense that the freest procedural family is still the best practical default.

Within the present rich-model hierarchy, the recommended family is **ring_hold**. It leads the executed frontier at 96 ns, 240 ns and 496 ns, preserves high QND quality, and is materially less fragile than the free procedural family under strong-drive stress. **procedural_segments** remains a useful specialist option when raw information fidelity is the dominant objective and the user is willing to accept a tighter QND budget. **nulling_tail** remains valuable when cavity emptying is the main concern, but it is no longer the overall frontier leader once transport and nonlinear backaction are both present.

The study goals were therefore met. The replay stack now includes explicit hardware transport and family-dependent non-QND disturbance, the practical frontier was re-ranked, the family comparison was validated, and a concrete control recommendation now exists.

7 Limitations and Future Work

7.1 Known Limitations

The strongest limitation is model fidelity. The effective mixing channel is intentionally phenomenological, constructed to expose the control consequences of strong-drive QND breakdown without claiming a microscopic derivation. That is sufficient for ranking trends, but it is not sufficient for quantitative hardware signoff.

The second limitation is coverage. The stable saved frontier only spans 96 ns, 240 ns and 496 ns, because the rich-model optimization was significantly more expensive and more fragile than the nominal study. That grid is enough to answer the main question, but it is not enough to map every duration crossover precisely.

The third limitation is the replay formalism. The public `cqed_sim` workflow still uses deterministic Lindblad replay rather than a stochastic measurement unraveling with time-dependent collapse structure. Repeated-readout quality is therefore quantified through matched replay metrics rather than trajectory-conditioned state update.

7.2 Suggested Improvements

- **[P1 | HIGH]** Calibrate the effective mixing layer against a more microscopic beyond-dispersive or experimentally informed model. The present ranking is plausible, but its absolute transition probabilities should not yet be treated as device-calibrated.
- **[P1 | MEDIUM]** Run a denser duration sweep in the 200 ns to 350 ns window. This is where the ranking changes from nominal `procedural_segments` dominance to rich-model `ring_hold` dominance.
- **[P2 | MEDIUM]** Introduce a constrained hybrid family built around `ring_hold` with a small number of additional smooth degrees of freedom. The goal is to test whether the remaining detector gap can be closed without reproducing the QND fragility of free procedural segments.
- **[P2 | HIGH]** Extend the study to a full three-mode replay or a more explicit readout-plus-storage validation layer once compute permits. The present conclusion rests on the two-mode-plus-mixing hierarchy.
- **[P3 | MEDIUM]** Add trajectory-based repeated-readout diagnostics when the simulator supports them. That would better separate detector noise, state disturbance, and readout-conditioned collapse.

7.3 Open Questions

The most important unresolved question is whether a lightly augmented smooth family can match the information performance of `procedural_segments` without inheriting its QND-stress penalty. A second open question is whether the nulling-tail family loses the frontier mainly because finite bandwidth destroys exact cancellation or because the richer model rewards smooth occupancy trajectories more than exact emptying per se. Finally, the relative roles of occupancy-driven versus slew-driven disturbance remain only partially separated in the current phenomenological model.

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